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Ordinary differential equations

In Chapter, 1 we discussed ordinary differential equations and, in particular, the hypergeometric equation

$$z(1-z)\frac{d^2 f}{dz^2} + (c - (1+a+b)z)\frac{df}{dz} - abf = 0. \quad (15.1)$$

We will denote this equation by $HG_{a,b,c}(z)$. (This notation is not very logical, but will be useful in a moment.) In this chapter we take this discussion further, fitting it into some of the theory developed in the book.

15.1 Conformal mapping

Here we take the parameters a, b, c to be real. We write

$$\alpha = (1 - c), \quad \beta = (b - a), \quad \gamma = c - a - b.$$

For simplicity, suppose $0 < \alpha, \beta, \gamma < 1$. We define a *curvilinear triangle* (Figure 15.1) to be a domain in $\mathbf{CP}^1$ whose boundary consists of three arcs of circlines.

Theorem 29. *If F, G are two independent solutions of the hypergeometric equation in the upper half-plane H , then the ratio G/F maps H to a curvilinear triangle. The points $z = 0, \infty, 1$ map to the vertices of the triangle, and the angles at the vertices are $\pi\alpha, \pi\beta, \pi\gamma$, respectively. Conversely, any such triangle can be obtained in this way.*

We will give two proofs, the first elementary (and focused on the first statement in the theorem) and the second more sophisticated (and focused on the second, converse statement). To begin, notice that it suffices to prove the result for *some* pair of solutions F, G . For if $\tilde{F}, \tilde{G}$ are another pair, we can write

$$\tilde{F} = pF + qG, \quad \tilde{G} = rF + sG$$

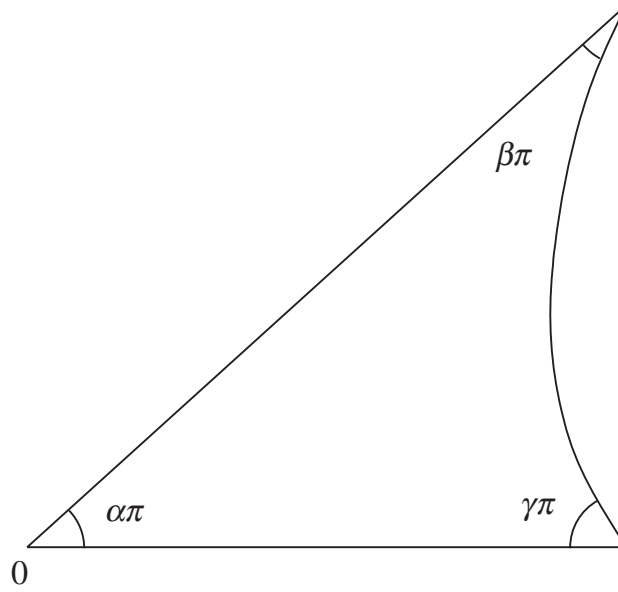


Figure 15.1 *The curvilinear triangle*

for constants p, q, r, s , and then $\tilde{G}/\tilde{F}$ is the composite of G/F and the Möbius map

$$w \mapsto \frac{pw + q}{rw + s}.$$

This uses the fact that Möbius maps preserve the class of curvilinear triangles, and the angles at vertices.

Back in Chapter 1, we mentioned that any second-order ordinary differential equation (ODE) with three regular singular points can be transformed to a hypergeometric equation by the action of Möbius maps and multiplication of the variable by a fixed function (Exercise 6 in Chapter 1). By the same token, the Möbius maps which permute the three points $0, 1, \infty$ transform one hypergeometric equation to another. Thus any property we can establish for solutions around 0 , say, gives a corresponding property for solutions around 1 and around ∞ .

To examine this symmetry, consider the Möbius map $T(z) = 1/(1 - z)$. This preserves the upper half-plane and permutes $0, 1, \infty$ cyclically. Write $w = T(z)$ and suppose that f satisfies the hypergeometric equation $HG_{a,b,c}(w)$, using notation in the obvious way. Rewrite this equation in terms of the variable z using the chain rule:

$$\frac{d}{dw} = (1 - z)^2 \frac{d}{dz}.$$

We get

$$\begin{aligned} & \frac{1}{1-z} \left(\frac{-z}{z-1} \right) (1-z)^2 \frac{d}{dz} \left((1-z)^2 \frac{df}{dz} \right) \\ & + \left(c - \frac{(1+a+b)}{1-z} \right) (1-z)^2 \frac{df}{dz} - abf = 0. \end{aligned}$$

This simplifies to

$$z(1-z) \frac{d^2 f}{dz^2} + P \frac{df}{dz} + \frac{ab}{1-z} f = 0,$$

where $P = 1 + a + b - c(1-z) - 2z$. Now write $f = (1-z)^r \tilde{f}$, where r is a constant to be determined. We get a second-order equation for $\tilde{f}$ of the form

$$z(1-z) \frac{d^2 \tilde{f}}{dz^2} + \tilde{P} \frac{d\tilde{f}}{dz} + \tilde{Q} \tilde{f} = 0. \quad (15.2)$$

A little calculation gives

$$\tilde{Q} = \frac{z(r^2 + r - cr) + ab - (1+a+b)r + cr}{1-z}.$$

To get an equation in hypergeometric form, we need the numerator to be a multiple of $1-z$, which is the condition $r^2 + r = (1+a+b)r - ab$ and hence $r^2 - (a+b)r + ab = 0$. Thus we need to take $r = a$ or b . It turns out to be convenient to choose $r = b$. With this choice,

$$\tilde{P} = (1+a+b-c) - z(1+b-c), \quad \tilde{Q} = -b(1+b-c),$$

so equation (15.2) is a hypergeometric equation with parameters a', b', c' , where $c' = 1+a+b-c$, $a' + b' = b+1-c$, $a'b' = b(1+b-c)$. We have another choice, of which parameter we call a' and which b' . The convenient choice is

$$a' = 1+b-c, \quad b' = b.$$

So, to sum up, we have shown that a function $\tilde{f}(z)$ satisfies $HG_{a',b',c'}(z)$ if and only if $w^{-b} \tilde{f}((w-1)/w)$ satisfies $HG_{a,b,c}(w)$. If we take the equivalent choice of parameters α, β, γ and α', β', γ' , the situation is much clearer, since we have

$$\alpha' = 1 - c' = c - a - b = \gamma,$$

$$\beta' = a' - b' = 1 - c = \alpha,$$

$$\gamma' = c' - a' - b' = a - b = \beta.$$

So, as T cyclically permutes the three points $0, 1, \infty$, we cyclically permute the parameters α, β, γ in the hypergeometric equation. (This is the reason for the particular choices made above—other choices would introduce signs.)

Recall the hypergeometric series

$$F(a, b, c; z) = 1 + \frac{ab}{1!c}z + \frac{a(a+1)b(b+1)}{2!c(c+1)}z^2 + \dots \quad (15.3)$$

This gives a holomorphic solution of equation (15.2) around 0. We denote this by F_0 . By straightforward calculation, we find that a second solution is given by

$$G_0(z) = z^{1-c} F(a-c+1, b-c+1, 2-c; z).$$

Here the power z^{1-c} is defined by cutting the plane along a ray in the lower half-plane. Now F_0 is real on the real axis. The second solution G_0 is real on the positive real axis but maps the negative real axis to the ray defined by the condition $\arg z = \pi\alpha$. So it is clear that G_0/F_0 maps the ‘semi-disc’ $\{z : |z| < 1, \operatorname{Im} z > 0\}$ to the wedge-shaped region $\{w : 0 < \arg w < \pi\alpha\}$. This map is illustrated in Figure 15.2. below. Now we apply our cyclic symmetry. We see that there are two solutions, F_1 and G_1 say, of the hypergeometric equation defined near 1 whose ratio maps a semi-disc in the upper half-plane to a wedge-shaped region with angle $2\pi\beta$. On the other hand, we can analytically continue F_0, G_0 to this neighbourhood of 1, and their ratio gives a map which is the composite of G_1/F_1 with a Möbius map. Similarly for the point at infinity. What we conclude is that if we extend F_0, G_0 over the upper half-plane by analytic continuation and set $f = G_0/F_0$, then f maps these three semi-discs to three wedge-shaped regions bordered by segments of circlines. Now it is clear (by thinking of the analytic continuation process along the real axis) that the borders of these wedge-shaped regions must match up. That is, f maps H to a curvilinear triangle. Now the derivative of f is W/F_0^2 , where $W = G'_0 F_0 - G_0 F'_0$ and W satisfies a first-order ODE

$$z(1-z)W' + (c - (1+a+b)z)W = 0.$$

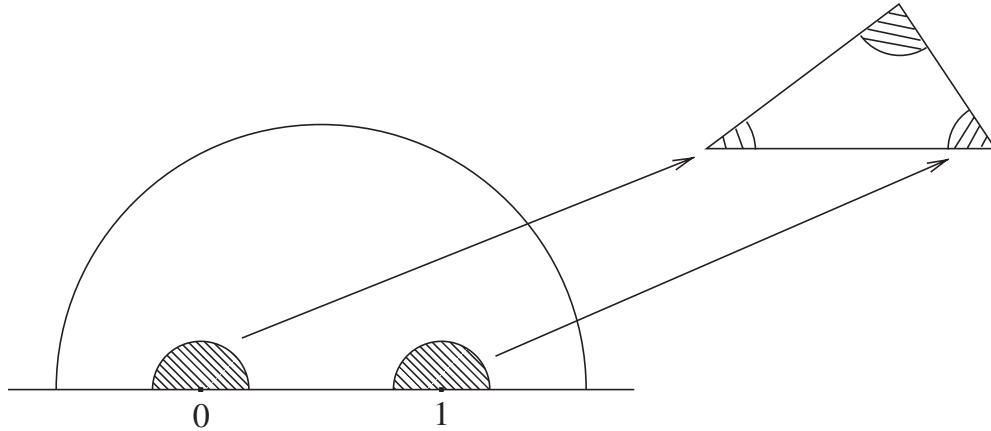


Figure 15.2 Map to a curvilinear triangle

This implies that W has no zeros in the upper half-plane. Hence the derivative of f does not vanish, and a straightforward topological argument shows that f maps H bijectively to the curvilinear triangle.

We can pin down exactly which curvilinear triangle is the image of the upper half-plane under the map $f = F/G_0$. We have that $f(0) = 0$, $f(1) = r_1$, $f(\infty) = r_2 e^{i\pi\alpha}$ for r_1, r_2 real and positive. Now elementary geometry determines the ratio r_2/r_1 from the angles:

$$\frac{r_2}{r_1} = \frac{\cos \pi\alpha + \cos \pi(\alpha + \beta) - \sin \pi\gamma + \sin \pi(\alpha + \beta)}{\cos \pi\alpha(\cos \pi\gamma + \cos \pi(\alpha + \beta)) - \sin \pi\alpha(\sin \pi\gamma - \sin \pi(\alpha + \beta))}.$$

On the other hand, at $w = 1$, the power series representations of F, G_0 still converge to give (Nehari 1975, Whittaker and Watson 1996)

$$f(1) = \frac{\Gamma(2-c)\Gamma(c-a)\Gamma(c-b)}{\Gamma(c)\Gamma(1-a)\Gamma(1-b)}.$$

Using this, one can write down explicitly the relation between the different local solutions of the hypergeometric equation arising from analytic continuation or, what is equivalent, the monodromy of the ODE with respect to the basis F, G_0 of local solutions. We refer to the literature for further details.

For the second point of view, we start our discussion by considering a Riemann surface S with a hyperbolic metric. Then we have a sheaf on S consisting of local Killing fields. In the upper half-plane, these are the vector fields $(a + 2bz + cz^2)(\partial/\partial z)$ for $a, b, c \in \mathbf{R}$. We call such a vector a ‘translation vector field’ if $ac = b^2$. This is the condition that the vector field has just one zero, which lies on the circle at infinity of the hyperbolic plane. We can define the same notion on a general hyperbolic surface: a translation vector field on some small open set $U \subset S$ is one which takes the above shape when we identify U with an open set in the upper half-plane. We may also consider ‘complexified translation vector fields’, for which we simply allow a, b, c to be complex with $ac = b^2$. (This complexification is a little unnatural from the hyperbolic-geometry point of view but will simplify our discussion. See the remarks on ‘Möbius structures’ below.)

Now suppose that we have a line bundle $L \rightarrow S$ which is a ‘square root’ of the tangent bundle in the sense that we have a holomorphic isomorphism $L \otimes L = TS$. We will write $L = TS^{1/2}$. (In fact, such square roots always exist, but are not generally unique.) In a local co-ordinate z , we can write sections of $TS^{1/2}$ as $f(\partial/\partial z), z^{1/2}$ and such a section has a square which is a vector field. In particular, on the upper half-plane, the translation vector fields are just those which are the squares of $(\alpha + \beta z)(\partial/\partial z)z^{1/2}$. So the following two conditions for a section $s = f(\partial/\partial z)z^{1/2}$,

- the square of f is a translation vector field,
- $d^2 f/dz^2 = 0$,

are equivalent. But the first condition makes sense on a general hyperbolic surface, so we conclude that, given a hyperbolic surface and a square root $TS^{1/2}$, *there is a natural second-order ODE for sections of $TS^{1/2}$* . To see this more explicitly, suppose we have a local holomorphic co-ordinate w on our surface which differs from z by composing with a holomorphic map $z = \phi(w)$. Then a section $f(z)(\partial/\partial z)^{1/2}$ in one co-ordinate transforms to $f(\phi(w))\phi'(w)^{-1/2}(\partial/\partial w)^{1/2}$. One finds that the simple equation $d^2 f/dz^2 = 0$ goes over to a condition on a section $F(w)(\partial/\partial w)^{1/2}$ which is the ODE

$$\frac{d^2 F}{dw^2} - QF = 0 \quad (15.4)$$

with

$$Q = \left(\psi^{1/2} \frac{d^2 \psi^{1/2}}{dw^2} \right),$$

where $\psi = d\phi/dw$. In turn, if we write $F = \lambda G$ for a fixed function $\lambda(w)$, this ODE transforms to the general form $d^2 G/dw^2 + \tilde{P} dG/dw + \tilde{Q}G = 0$ for $G(w)$.

Another point of view on this is to start with a Riemannian metric, written as $\rho^2 |dw|^2$. The Gauss curvature is $\rho^{-2} \Delta \log \rho$, so the condition that the metric is hyperbolic is $\Delta \log \rho = -\rho^2$. The uniqueness of the hyperbolic metric implies that we can find a function $z = \phi(w)$ as above, so that $\rho = |\phi'|/\text{Im } \phi$. Then we can define Q by the above formula, and hence the ODE (15.4) for $F(z)(\partial/\partial w)^{1/2}$. The surprising thing is that this function Q is independent of the choice of ϕ and, when interpreted in terms of sections of $TS^{1/2}$, the ODE is independent of the choice of local co-ordinate.

We next introduce the concept of a hyperbolic surface with *cone singularities*. We write the metric on the hyperbolic plane in geodesic co-ordinates as $dr^2 + \sinh^2 r d\theta^2$. Given a *cone angle* $2\pi\alpha$, we define another metric on the punctured plane to be $dr^2 + \alpha^2 \sinh^2 r d\theta^2$. This has a singularity at the origin but away from the origin is hyperbolic: locally, we can just take $(r, \alpha\theta)$ as a new coordinate. We define a hyperbolic surface with cone singularities to be a smooth surface with a hyperbolic metric defined on the complement of a finite subset which agrees with this local model (for an appropriate choice of cone angle) around each of these points. Given an orientation, we get a genuine Riemann surface structure extending across the cone points. From another point of view, we could make the definition that we have a Riemann surface with a singular metric which, around the singular points, can be written in the standard form

$$\frac{\alpha^2 |z|^{2(\alpha-1)}}{(1 - |z|^{2\alpha})^2} |dz|^2,$$

which is what one obtains from the disc model of the hyperbolic plane by making the change of variable $w = z^\alpha$.

Now suppose we have a compact hyperbolic Riemann surface with cone singularities and a choice of $TS^{1/2}$. Then, away from the singularities, we get an ODE for sections of $TS^{1/2}$ as above. In a local co-ordinate around a singular point, this has the form $F'' - QF = 0$ for a section $F(w)(\partial/\partial w)^{1/2}$. Then Q is defined on a punctured disc, and the key point is that it has *at worst a double pole*. Thus the ODE has a regular singular point. This is simply a matter of checking in the explicit local model for a cone singularity.

Now return to a triangle T in the Riemann sphere and suppose that this is actually a hyperbolic triangle in the upper half-plane. We take two copies of T and glue the edges to construct a hyperbolic surface S with cone singularities. The cone angles are $2\pi\alpha, 2\pi\beta, 2\pi\gamma$. The underlying Riemann surface is homeomorphic to the sphere, so we know that it is isomorphic to the standard sphere and we can fix an isomorphism taking $0, 1, \infty$ to the three singular points. By symmetry, the isomorphism must map one half-plane to the original triangle T and, with appropriate choices of orientation, we can suppose that this is the upper half-plane.

Now we define a square root of the tangent bundle of the Riemann sphere as follows. The section $\partial/\partial w$ of the tangent bundle has a double zero at the point at infinity, and we decree that local holomorphic sections of $(TCP^1)^{1/2}$ are sections $f(\partial/\partial w)^{1/2}$, where f has at worst a simple pole at infinity. So we have an ODE for sections $F(w)(\partial/\partial w)^{1/2}$ of the form $F'' - QF = 0$, where the only singularities of Q are the three points $0, 1, \infty$ and the singularities are regular, in local co-ordinates around these points. We know that any second-order ODE on the Riemann sphere with three regular singular points can be transformed to a hypergeometric equation, and it follows from this that the map from the upper half-plane to the triangle T is given by a ratio of solutions of such an equation. In more detail, the ODE must have the form $F'' = QF$, where Q has at worst double poles at $0, 1$. The regularity condition at infinity is that $Q(w) = O(w^{-1})$ as $w \rightarrow \infty$, so we must have

$$Q(w) = A(w^{-1} - (w - 1)^{-1}) + Bw^{-2} + C(w - 1)^{-2}.$$

Now, writing $F = \lambda G$ for an appropriate (multivalued) function of the form $\lambda(w) = w^p(w - 1)^q$, one can transform this ODE to the standard hypergeometric form. A local analysis around the singular points relates the cone angles to the indicial roots and hence to the parameters a, b, c , but of course we have already know how this must go from our preceding discussion.

An advantage of this second point of view is that it extends to a general ‘curvilinear polygon’ Π . Taking the double of Π and arguing as above, we see that the conformal map from the upper half-plane to Π is given by the ration of two solutions of an ODE $F'' + QF$, where

$$Q(w) = \sum a_i(z - z_i)^{-2} + b_i(z - z_i)^{-1},$$

and a_i, b_i, z_i are real. We also need $\sum a_i = 0$ to achieve a regular singularity at infinity. Knowledge of the angles of Π gives some constraints on the parameters, although it is not known how to determine these explicitly from Π .

The natural context for the discussion above is not really hyperbolic geometry but the more general notion of *Möbius structure*. A sign of this is the rather unnatural complexification we introduced in the discussion above. A Möbius structure is given by a smooth surface with an atlas of charts which compare by elements of $PSL(2, \mathbb{C})$, acting on domains of the Riemann sphere. From this point of view, a hyperbolic structure is the special case when the maps come from $PSL(2, \mathbb{R}) \subset PSL(2, \mathbb{C})$ and the domains all lie in the upper half-plane. Given a Möbius structure, we have a subsheaf of a holomorphic vector field, modelled on those on the Riemann sphere, and the whole theory works as before.

Remark

1. Notice that cone angles of $2\pi/r$, for integer r , arise very naturally when we consider quotients. For example, the quotient of H by $PSL(2, \mathbb{Z})$ has a hyperbolic metric with two cone points having angles $\pi, 2\pi/3$, reflecting the fact that $PSL(2, \mathbb{Z})$ does not act freely.
2. The choice of a square root of the tangent bundle is related to lifting the homomorphism $\pi_1 \rightarrow PSL(2, \mathbb{R})$, which we have for a hyperbolic surface, to a homomorphism $\pi_1 \rightarrow SL(2, \mathbb{R})$.

15.2 Periods of holomorphic forms and ordinary differential equations

15.2.1 The hypergeometric equation

Consider the function of a complex variable λ

$$F(\lambda) = 2 \int_0^\lambda \frac{dz}{\sqrt{z(z-1)(z-\lambda)}} dz.$$

We can consider this as defined for λ in the upper half-space, say, and the integration path to be a line segment, on which we make a choice of definition of the square root. We can also regard $F(\lambda)$ as the integral around a closed loop γ on the Riemann surface associated to the equation $w^2 = z(z-1)(z-\lambda)$ of the holomorphic 1-form

$$\theta = \frac{dz}{\sqrt{z(z-1)(z-\lambda)}}.$$